

# Employee\_Sales — Complete Descriptive Statistics

Manual line-by-line calculation • No shortcuts

## Dataset

42, 45, 47, 49, 50, 52, 53, 55, 56, 58, 60, 62, 64, 65, 68, 70, 72, 75, 78, 82, 58

**Number of observations:**  $n = 21$

## Statistics covered

Count, Sum, Mean, Median, Mode, Minimum, Maximum, Range, Sample Variance, Standard Deviation, Standard Error, Skewness, Kurtosis, 95% Confidence Level, and 95% Confidence Interval.

## 1. COUNT — Line by Line

Observation 1: 42

Observation 2: 45

Observation 3: 47

Observation 4: 49

Observation 5: 50

Observation 6: 52

Observation 7: 53

Observation 8: 55

Observation 9: 56

Observation 10: 58

Observation 11: 60

Observation 12: 62

Observation 13: 64

Observation 14: 65

Observation 15: 68

Observation 16: 70

Observation 17: 72

Observation 18: 75

Observation 19: 78

Observation 20: 82

Observation 21: 58

There are 21 observations.

**Count = 21**

## 2. SUM — Line by Line

Step 1:  $0 + 42 = 42$

Step 2:  $42 + 45 = 87$

Step 3:  $87 + 47 = 134$

Step 4:  $134 + 49 = 183$

Step 5:  $183 + 50 = 233$

Step 6:  $233 + 52 = 285$

Step 7:  $285 + 53 = 338$

Step 8:  $338 + 55 = 393$

Step 9:  $393 + 56 = 449$

Step 10:  $449 + 58 = 507$

Step 11:  $507 + 60 = 567$

Step 12:  $567 + 62 = 629$

Step 13:  $629 + 64 = 693$

Step 14:  $693 + 65 = 758$

Step 15:  $758 + 68 = 826$

Step 16:  $826 + 70 = 896$

Step 17:  $896 + 72 = 968$

Step 18:  $968 + 75 = 1043$

Step 19:  $1043 + 78 = 1121$

Step 20:  $1121 + 82 = 1203$

Step 21:  $1203 + 58 = 1261$

**Sum = 1261**

### 3. MEAN — Line by Line

Formula: Mean = Sum ÷ Count

Mean =  $1261 \div 21$

Mean = 60.04761905

**Mean = 60.04761905**

## 4. MEDIAN — Line by Line

Step 1: Arrange the values from smallest to largest.

42, 45, 47, 49, 50, 52, 53, 55, 56, 58, 58, 60, 62, 64, 65, 68, 70, 72, 75, 78, 82

Step 2: Count = 21 (an odd number).

Step 3: Median position =  $(n + 1) \div 2$

Median position =  $(21 + 1) \div 2 = 11$

Step 4: The 11th value is 58.

**Median = 58**

## 5. MODE — Line by Line

Check how many times each value occurs.

| Value | Frequency |
|-------|-----------|
| 42    | 1         |
| 45    | 1         |
| 47    | 1         |
| 49    | 1         |
| 50    | 1         |
| 52    | 1         |
| 53    | 1         |
| 55    | 1         |
| 56    | 1         |
| 58    | 2         |
| 60    | 1         |
| 62    | 1         |
| 64    | 1         |
| 65    | 1         |
| 68    | 1         |
| 70    | 1         |
| 72    | 1         |
| 75    | 1         |
| 78    | 1         |
| 82    | 1         |

58 occurs 2 times. Every other value occurs 1 time.

**Mode = 58**

## 6. MINIMUM, MAXIMUM & RANGE

Minimum = smallest value = 42

Maximum = largest value = 82

Range = Maximum – Minimum

Range = 82 – 42

**Range = 40**

## 7. SAMPLE VARIANCE — Every Row

Mean = 60.04761905

Formula: Sample Variance =  $\Sigma(X - \bar{X})^2 \div (n - 1)$

| X     | X – Mean     | (X – Mean) <sup>2</sup> |
|-------|--------------|-------------------------|
| 42    | -18.04761905 | 325.71655329            |
| 45    | -15.04761905 | 226.43083900            |
| 47    | -13.04761905 | 170.24036281            |
| 49    | -11.04761905 | 122.04988662            |
| 50    | -10.04761905 | 100.95464853            |
| 52    | -8.04761905  | 64.76417234             |
| 53    | -7.04761905  | 49.66893424             |
| 55    | -5.04761905  | 25.47845805             |
| 56    | -4.04761905  | 16.38321995             |
| 58    | -2.04761905  | 4.19274376              |
| 60    | -0.04761905  | 0.00226757              |
| 62    | 1.95238095   | 3.81179138              |
| 64    | 3.95238095   | 15.62131519             |
| 65    | 4.95238095   | 24.52607710             |
| 68    | 7.95238095   | 63.24036281             |
| 70    | 9.95238095   | 99.04988662             |
| 72    | 11.95238095  | 142.85941043            |
| 75    | 14.95238095  | 223.57369615            |
| 78    | 17.95238095  | 322.28798186            |
| 82    | 21.95238095  | 481.90702948            |
| 58    | -2.04761905  | 4.19274376              |
| TOTAL |              | 2486.95238095           |

Step 1: Sum of squared deviations = 2486.95238095

Step 2:  $n - 1 = 21 - 1 = 20$

Step 3: Variance =  $2486.95238095 \div 20$

**Sample Variance = 124.347619048**



## 8. STANDARD DEVIATION — Line by Line

Formula: Standard Deviation =  $\sqrt{\text{Sample Variance}}$

SD =  $\sqrt{124.347619048}$

SD = 11.151126358

**Standard Deviation = 11.15112636**

## 9. STANDARD ERROR — Line by Line

Formula: Standard Error = Standard Deviation  $\div$   $\sqrt{n}$

$$SE = 11.15112636 \div \sqrt{21}$$

$$\sqrt{21} = 4.58257569$$

$$SE = 11.15112636 \div 4.58257569$$

$$\text{Standard Error} = 2.433375268$$

## 10. SKEWNESS — Line by Line

Excel's sample skewness formula:

$$\text{Skewness} = [n / ((n-1)(n-2))] \times \Sigma[(X - \bar{X})/s]^3$$

n = 21,  $\bar{X}$  = 60.04761905, s = 11.15112636

| X     | X-Mean    | (X-Mean)/SD | z <sup>3</sup> |
|-------|-----------|-------------|----------------|
| 42    | -18.04762 | -1.618457   | -4.23939160    |
| 45    | -15.04762 | -1.349426   | -2.45723760    |
| 47    | -13.04762 | -1.170072   | -1.60190808    |
| 49    | -11.04762 | -0.990718   | -0.97241099    |
| 50    | -10.04762 | -0.901041   | -0.73153190    |
| 52    | -8.04762  | -0.721687   | -0.37587723    |
| 53    | -7.04762  | -0.632010   | -0.25244749    |
| 55    | -5.04762  | -0.452656   | -0.09274778    |
| 56    | -4.04762  | -0.362978   | -0.04782365    |
| 58    | -2.04762  | -0.183624   | -0.00619143    |
| 60    | -0.04762  | -0.004270   | -0.00000008    |
| 62    | 1.95238   | 0.175084    | 0.00536707     |
| 64    | 3.95238   | 0.354438    | 0.04452667     |
| 65    | 4.95238   | 0.444115    | 0.08759633     |
| 68    | 7.95238   | 0.713146    | 0.36268978     |
| 70    | 9.95238   | 0.892500    | 0.71092660     |
| 72    | 11.95238  | 1.071854    | 1.23142246     |
| 75    | 14.95238  | 1.340885    | 2.41087586     |
| 78    | 17.95238  | 1.609916    | 4.17263076     |
| 82    | 21.95238  | 1.968625    | 7.62937006     |
| 58    | -2.04762  | -0.183624   | -0.00619143    |
| TOTAL |           |             | 5.87164633     |

$$\Sigma z^3 = 5.87164633$$

$$n / ((n-1)(n-2)) = 21 / (20 \times 19) = 0.05526316$$

$$\text{Skewness} = 0.05526316 \times 5.87164633$$

$$\text{Skewness} = 0.324485718$$

Interpretation: positive, so the distribution is slightly right-skewed.

## 11. KURTOSIS — Line by Line

Excel's KURT gives excess kurtosis:

$$\text{Kurtosis} = [n(n+1)/((n-1)(n-2)(n-3))] \times \sum z^4 - [3(n-1)^2/((n-2)(n-3))]$$

n = 21,  $\bar{X} = 60.04761905$ , s = 11.15112636

| X     | z = (X-Mean)/SD | z <sup>4</sup> |
|-------|-----------------|----------------|
| 42    | -1.618457       | 6.86127321     |
| 45    | -1.349426       | 3.31586014     |
| 47    | -1.170072       | 1.87434756     |
| 49    | -0.990718       | 0.96338485     |
| 50    | -0.901041       | 0.65914004     |
| 52    | -0.721687       | 0.27126558     |
| 53    | -0.632010       | 0.15954924     |
| 55    | -0.452656       | 0.04198279     |
| 56    | -0.362978       | 0.01735895     |
| 58    | -0.183624       | 0.00113690     |
| 60    | -0.004270       | 0.00000000     |
| 62    | 0.175084        | 0.00093969     |
| 64    | 0.354438        | 0.01578193     |
| 65    | 0.444115        | 0.03890283     |
| 68    | 0.713146        | 0.25865076     |
| 70    | 0.892500        | 0.63450203     |
| 72    | 1.071854        | 1.31990526     |
| 75    | 1.340885        | 3.23270791     |
| 78    | 1.609916        | 6.71758660     |
| 82    | 1.968625        | 15.01936510    |
| 58    | -0.183624       | 0.00113690     |
| TOTAL |                 | 41.40477827    |

$$\sum z^4 = 41.40477827$$

$$\text{First factor} = 21 \times 22 / (20 \times 19 \times 18) = 0.06754386$$

$$\text{Second term} = 3 \times 20^2 / (19 \times 18) = 3.50877193$$

$$\text{Kurtosis} = 0.06754386 \times 41.40477827 - 3.50877193$$

**Excess Kurtosis = -0.712133398**

Interpretation: negative excess kurtosis means a relatively flatter/lighter-tailed distribution than normal.

## 12. 95% CONFIDENCE LEVEL — Line by Line

Excel's 'Confidence Level (95.0%)' is the margin of error, not the two interval endpoints.

Formula: Margin of Error = t critical × Standard Error

Degrees of freedom =  $n - 1 = 21 - 1 = 20$

For 95% confidence and df = 20: t critical  $\approx 2.085963447$

Standard Error = 2.433375268

Margin =  $2.085963447 \times 2.433375268$

**95% Margin of Error = 5.075931861**

## 13. 95% CONFIDENCE INTERVAL

Formula: Mean  $\pm$  Margin of Error

Lower =  $60.04761905 - 5.075931861 = 54.971687187$

Upper =  $60.04761905 + 5.075931861 = 65.123550909$

**95% Confidence Interval = (54.971687187, 65.123550909)**

## 14. FINAL ANSWERS

| Statistic              | Result                       |
|------------------------|------------------------------|
| Count                  | 21                           |
| Sum                    | 1261                         |
| Mean                   | 60.04761905                  |
| Median                 | 58                           |
| Mode                   | 58                           |
| Standard Deviation     | 11.15112636                  |
| Sample Variance        | 124.34761905                 |
| Kurtosis               | -0.712133398                 |
| Skewness               | 0.324485718                  |
| Range                  | 40                           |
| Minimum                | 42                           |
| Maximum                | 82                           |
| Standard Error         | 2.433375268                  |
| Confidence Level (95%) | 5.075931861                  |
| 95% CI                 | (54.971687187, 65.123550909) |

Important note: The calculations above use the 21 values exactly as supplied, including 58 appearing twice. This reproduces the Excel-style variance, standard deviation, standard error, skewness, kurtosis, and 95% confidence-level approach.